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Urban planning in the age of large language models: Assessing OpenAI o1's performance and capabilities across 556 tasks

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ABSTRACT

Integrating Large Language Models (LLMs) into urban planning presents significant opportunities to enhance efficiency and support data-driven city development strategies. Despite their potential, the specific capabilities of LLMs within the urban planning context remain underexplored, and the field lacks standardized benchmarks for systematic evaluation. This study presents the first comprehensive evaluation focused on OpenAI o1's performance and capabilities in urban planning, systematically benchmarking it against GPT-3.5 and GPT-4o using an original open-source benchmark comprising 556 tasks across five critical categories: urban planning documentation, examinations, routine data analysis, AI algorithm support, and thesis writing. Through rigorous testing and manual analysis of 170,627 words of generated output, OpenAI o1 consistently outperformed its counterparts, achieving an average performance score of 84.08 % compared to 69.30 % for GPT-4o and 45.27 % for GPT-3.5. Our findings highlight o1's strengths in domain knowledge mastery, basic operational competence, and coding capabilities, demonstrating its potential applications in information retrieval, urban data analytics, planning decision support, educational assistance, and LLM-based agent development. However, significant limitations were identified, including inability in urban design, susceptibility to fabricating information, moderate academic writing quality, challenges in high-level professional examinations, and spatial reasoning, and limited support for specialized or emerging AI algorithms. Future optimizations should prioritize enhancing multimodal integration, implementing robust validation mechanisms, adopting adaptive learning strategies, and enabling domain-specific fine-tuning to meet urban planners' specialized needs. These advancements would enable LLMs to better support the evolving demands of urban planning, allowing professionals to focus more on strategic decision-making and the creative aspects of the field.

1. Introduction

Urban planning, also known as city or town planning, is both a technical and political process aimed at developing and organizing the physical form, economic functions, social systems, and transportation networks of urban spaces. The goal is to create functional, sustainable, efficient, and aesthetically pleasing environments while balancing the often competing demands of multiple stakeholders, including politicians, governments, financiers, developers, the public, architects, and engineers (Jiang et al., 2024; Rittel & Webber, 1973). The core objective of urban planning is widely recognized as "shaping better places for people" (Carmona, 2021).

Cities, as complex systems, share many characteristics with natural systems: they are open, intricate, and self-organizing (Allen, 2012). A primary challenge in urban planning lies in navigating this complexity and the interconnectivity of urban systems. Urban designers must address existing site issues, anticipate future trends, and make strategic decisions that balance diverse and often conflicting interests. This requires a diverse set of skills and knowledge, including problem formulation, data collection and analysis, conceptual design, drawing representation, policy formulation, stakeholder engagement, and project management. Additionally, integrating expertise from various

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fields—such as architecture, law, economics, politics, environmental science, and geography—is essential to the planning process. However, many labor-intensive tasks within the traditional planning workflow—such as manual site analysis, data processing, and navigating dense regulatory documents—are often seen as inefficient and time-consuming, relying heavily on planners' experience and qualitative judgment (Andrews et al., 2022).

In recent years, the widespread availability of big data, enhanced computing capabilities, and breakthroughs in Artificial Intelligence (AI) techniques have revolutionized urban planning. Unlike manual tasks, AI can efficiently handle high-volume activities, automate repetitive tasks, learn from large datasets, and adapt to new patterns and behaviors (F. Xu, Zhang, Gao, Feng, & Li, 2023; Zhang, Pee, Pan, & Liu, 2022). AI has become a transformative tool for improving the efficiency of urban planning processes and generating data-driven insights to tackle urban complexities (Andrews et al., 2022; Son et al., 2023; F. Xu et al., 2023). Research highlights a broad range of prospective AI applications in urban studies, spanning social, economic, environmental, and technological domains. These include generative urban design (Jiang et al., 2024), land-use planning (Chaturvedi & de Vries, 2021), urban form planning (Koumetio Tekouabou, Diop, Azmi, & Chenal, 2023), social sensing (Biljecki & Ito, 2021; Zhao, Lu, Huang, & Lin, 2024), urban renewal/regeneration (He, Zhang, Yao, & Li, 2023), traffic system management (Sanchez, Shumway, Gordner, & Lim, 2023; Son et al., 2023), urban monitoring and development control (Son et al., 2023), zoning and permitting (Andrews et al., 2022), and climate change mitigation (Chen et al., 2023), among others. These applications leverage data science to generate a multitude of decision-making alternatives (Sanchez et al., 2023), enhancing city management and advancing smart, sustainable development goals (SDGs).

Among these applications, neural network algorithms have gained immense popularity in recent years. Following the introduction of AlexNet (Krizhevsky, Sutskever, & Hinton, 2012) in the early 2010s, neural networks paved the way for more advanced models such as VGG, ResNet, and, more recently, Transformer architectures. While these models excel as narrow experts, performing exceptionally well on specific tasks they are trained for (Biljecki & Ito, 2021; Chaturvedi & de Vries, 2021; Chen et al., 2023), they often struggle to generalize to the complex, real-world scenarios in urban planning, which typically involve diverse inputs from various stakeholders (Chowdhery et al., 2023; Radford et al., 2019).

The advent of Large Language Models (LLMs), equipped with billions of parameters and trained on vast datasets, has opened new horizons across various domains by providing advanced capabilities for language-related tasks such as question answering, summarization, reasoning, information retrieval, and coding (Brown et al., 2020; Kaur et al., 2024; Radford et al., 2019; L. Xu et al., 2024). Since the introduction of the game-changing GPT-3 (Brown et al., 2020), there has been an extraordinarily rapid evolution in the capabilities of LLMs, driven by increases in model size and the volume and quality of pre-training data (Chowdhery et al., 2023; S. Wu et al., 2024). This includes a growing number of LLMs from various companies (Yang et al., 2024), such as GPT-4o, and o1 from OpenAI, Gemini and Gemma from Google, LLaMA from Meta, Claude from Anthropic, ChatGLM from ZhipuAI, DeepSeek from DeepSeek AI, and Qwen from Alibaba. The size of these models has grown from 117 M parameters (Radford et al., 2019) to over 540B parameters (Chowdhery et al., 2023), continuing to push the state of the art forward.

Despite their impressive performance, LLMs still face challenges such as bias, limited generalizability, and difficulty handling tasks that require common sense or domain-specific knowledge (Gu et al., 2021). Moreover, LLMs may produce outputs that are plausible but incorrect, posing challenges for their application in critical decision-making contexts. Recent studies also suggest that LLMs trained on next token prediction pre-training always struggle with complex tasks involving logical reasoning (Xie et al., 2024).

In response to these challenges, OpenAI introduced its model o1 in September 2024, incorporating Chain-of-thought (CoT) prompting (Saunders et al., 2022) to improve inference by breaking tasks into intermediate steps that mirror human reasoning. Additionally, o1 integrates advanced reinforcement learning techniques to enhance its reasoning while ensuring alignment with human values. OpenAI's benchmarks show that o1 excels in tasks requiring structured reasoning, such as programming and problem-solving (Jaech et al., 2024). Recent research has also demonstrated o1's superior effectiveness compared to other LLMs in downstream tasks ranging from natural language processing (NLP) reasoning tasks (S. Wu et al., 2024; Zhong et al., 2024) to applications in medicine (Xie et al., 2024), mathematics (de Winter, Dodou, & Eisma, 2024), and education (Latif et al., 2024).

Given its enhanced reasoning, context retention, and task-specific adaption capabilities, OpenAI o1 holds significant potential to transform urban planning by improving existing planning processes, increasing operational efficiency, and allowing planners to focus more on the human factors of their practice (Andrews et al., 2022). Despite this promise, critical questions persist regarding o1's practical capabilities within the urban planning domain. The current absence of standardized benchmarks impedes systematic evaluation and obscures our understanding of model performance across diverse tasks. Consequently, thoroughly identifying the strengths and limitations of LLMs in professional practice, alongside outlining future developments to enhance their utility, represents a crucial yet under-examined step toward the responsible integration of these tools into urban planning workflows. This research directly addresses these critical gaps. By conducting a systematic evaluation of o1's performance against GPT-3.5 and GPT-4o across a comprehensive benchmark of urban planning tasks, this study aims to provide a detailed assessment of current LLM capabilities and limitations, thereby informing and guiding future advancements in LLM applications within the field. This study makes several significant contributions to the field:

1) We developed and released a comprehensive open-source benchmark comprising 556 tasks across five major categories: urban planning documentation, relevant professional examinations, routine data analysis, AI algorithm support, and thesis writing tasks.
2) For AI algorithm code generation tasks, we curated and provided datasets including numeric, text, image, and spatial data, enabling thorough testing of model training and algorithm implementation capabilities.
3) Through a detailed analysis of 170,627 words of output from OpenAI o1 and its comparison models (GPT-3.5 and GPT-4o), we offer valuable insights into the current capabilities and limitations of LLMs for urban planning applications.

These findings significantly contribute to the ongoing discourse on AI integration in urban planning. Furthermore, they establish a foundation for future research aimed at optimizing LLMs to better serve the specialized needs of urban planning professionals.

This paper is structured as follows: Section 2 reviews relevant benchmark studies on OpenAI o1 and the application of LLMs in city-related fields. Section 3 details the materials and methods, including LLM selection, evaluation strategy, and descriptions of the five task categories. Section 4 presents the test results, offering an overview of the LLMs' performance and detailed analyses for each task category. Section 5 discusses the study's significance, the potential applications of o1 in urban planning, its identified shortcomings, directions for future improvements, and the research's limitations. Finally, Section 6 concludes with key findings and their implications for AI in urban planning.

2. Related works[1]

2.1. Evaluating the effectiveness of OpenAI-o1 in specific tasks

Several studies have evaluated the effectiveness of OpenAI o1 across a variety of tasks, providing insights into its capabilities and limitations. Zhong et al. (2024) conducted a comprehensive evaluation of o1's performance across a wide range of reasoning tasks, spanning fields such as computer science, mathematics, medicine, and social sciences. Latif et al. (2024) examined o1's ability to engage in higher-order thinking across 14 distinct cognitive domains, utilizing specific datasets and instruments designed to measure critical thinking, systems thinking, design thinking, and others. S. Wu et al. (2024) compared o1's reasoning patterns with existing Test-time Compute methods across three core domains: mathematics, code, and commonsense reasoning. In the medical field, Xie et al. (2024) examined o1's performance in understanding, reasoning, and multilinguality, evaluating it on six different tasks using 37 diverse datasets.

Overall, these evaluations indicate that o1 generally outperforms other LLMs in terms of reasoning ability and performance across most benchmarks. However, it occasionally encounters errors on simpler problems and shows difficulty in tasks requiring high-level abstraction (Latif et al., 2024), as well as in spatially complex domains (K. Wang et al., 2024). Furthermore, its capacity to address highly specialized concepts in niche areas remains limited (Zhong et al., 2024).

2.2. City-related LLM-driven approaches

Researchers in urban planning-related fields like geography and transportation have increasingly attempted to equip LLMs with domain expertise through domain-specific fine-tuning. For instance, PlanGPT (Zhu et al., 2024) combines a custom embedding model with vector database retrieval to extract precise information from urban planning texts. This model, fine-tuned on a curated dataset and supported by PlanAgent, integrates network resources and domain tools for urban planning documentation tasks. CityGPT (F. Xu et al., 2023) began with continued pre-training of the Baichuan2-7B model using general and domain-specific text corpora to create CityGPT-base. Through supervised fine-tuning using general-purpose, domain-specific, and task-solving chat datasets, this model developed urban intelligence capabilities, enabling applications from embodied simulation agents for mobility and economic activities to decision support for location and scheduling recommendations. Similarly, Feng et al. (2024) fine-tuned ChatGLM3-6B on a dataset constructed for *CityQA*, *CityWalk,* and *CityReasoning* tasks to enhance the model's spatial cognition, quantitative reasoning, and understanding of urban elements and functions. Additional efforts have fine-tuned open-source LLMs such as ChatGLM (Huang et al., 2024; X. Wang et al., 2023), Vicuna (Li et al., 2024), and LLaMA (Deng et al., 2024) for various applications, including urban renewal knowledge QA (X. Wang et al., 2023), address standardization (Huang, Chen, et al., 2024), spatio-temporal prediction (Li et al., 2024), and geoscience understanding (Deng et al., 2024).

Another significant approach involves deploying multiple specialized LLM agents to collaborate on complex urban challenges. Zhou, Lin, Jin, and Li (2024) introduced a multi-agent framework using GPT-3.5 and GPT-4 to simulate resident discussions and urban planner interactions. This framework adapts site plans based on resident feedback, facilitating participatory urban planning and accommodating diverse stakeholder needs. Liang, Liu, Wang, and Zhao (2024) employed GPT-4 to summarize unstructured event details into structured summaries and integrate these with past mobility patterns. This enables predictions of human mobility under public event scenarios, which are challenging to model with traditional methods due to sparse historical data and event complexity. Similarly, Jiawei et al. (2024) proposed a GPT-3.5-based framework for generating individual daily activities based on real-world human activity data, providing a valuable tool for urban mobility analysis. Lastly, Jin and Ma (2024) introduced a GPT-4-based Parking Planning Agent. By combining structured prompts with pre-packaged analytical tools, this agent processes and evaluates parking facility data, optimizing solutions for scenarios where autonomous and human-driven vehicles coexist.

These advancements highlight the broad potential of LLMs in dynamic urban analysis and planning, fostering proactive decision-making and innovative solutions to complex urban challenges. In our study, some open-source datasets from these studies have been applied for our tests.

3. Materials and methods

3.1. Selected LLMs and core evaluation strategy

In this study, OpenAI o1 was selected as the primary LLM for assessing performance and capabilities in urban planning tasks. To provide robust benchmarks for this assessment, GPT-3.5 and GPT-4o were also selected for comparative evaluation. GPT-3.5, an advanced language model by OpenAI, is known for its enhanced contextual understanding and represents a milestone in LLM development due to its coherent and human-like responses, making it a benchmark for industry performance evaluations. GPT-4o, the successor to GPT-3.5, boasts significant improvements in reasoning and language comprehension, offering faster responses and achieving new performance heights.

Our evaluations employed a zero-shot approach, with no prompt tuning, to assess the models' initial, unaided performance. All interactions with the selected LLMs were executed in Python via their respective official Application Programming Interfaces (APIs). This approach allows us to observe the models' direct response capabilities to given urban planning tasks. For each task, we employed the same prompting strategies across the models. Two human evaluators specializing in both LLMs and the relevant urban planning tasks conducted the assessments. Both evaluators received comprehensive training to ensure uniform understanding and consistent application of task-specific scoring criteria. Any discrepancies arising from their initial independent evaluations were subsequently resolved through thorough discussion until a consensus was achieved.

3.2. Test task categories and details

One of the most critical tasks in urban planning involves conceptual design and drawing representation. Since current LLMs are not yet capable of effectively performing such spatially and visually intensive tasks, they were excluded from our evaluation. In total, five task categories were designed to assess LLMs' performance and capabilities in urban planning:

1) Urban planning documentation tasks: These include materials collection, text information extraction, and documentation generation. They represent the routine work of urban planners in gathering data, synthesizing information, and producing reports. These activities are fundamental to urban planning workflows and support decision-making processes.
2) Urban planning examination tasks: These include questions from the Master of Urban Planning entrance examination and the Certified Urban-Rural Planner examination. These tasks evaluate both foundational and professional knowledge in areas such as history, principles, regulations, and interdisciplinary concepts, which are essential for practitioners in the field.

[1] Our review of related works incorporates both peer-reviewed publications and significant pre-print articles (e.g., from arXiv) to comprehensively cover recent and rapidly evolving research in the field of LLMs and their applications.

3) Routine data analysis tasks: These include questions on GIS concepts, spatial reasoning, geospatial analysis, coding, mapping, and related topics. Urban planners regularly engage in analytical activities using tools such as GIS and programming languages (e.g., Python) to assess and simulate urban conditions. These interdisciplinary skills are critical for data-driven planning, enabling planners to interpret complex datasets and forecast urban trends.
4) AI algorithm tasks: This category tests the application of various AI algorithms to analyze numeric, spatial, textual, and image data related to the urban environment. It reflects the growing integration of AI in urban research and practice, where advanced algorithms help address complex planning challenges and extract insights from diverse data sources.
5) Thesis writing tasks: These tasks evaluate LLMs' ability to generate high-quality academic introductions based on research titles, highlights, abstracts, and other components of research papers. This skill is vital for planners engaged in research or required to produce high-quality, coherent, and well-referenced written outputs such as research articles, dissertations, or comprehensive reports.

Chosen to reflect the multifaceted demands of modern urban planning, these five categories form a comprehensive benchmark for assessing LLM proficiency across a range of essential activities, from foundational documentation and knowledge application to advanced data analytics and scholarly research. Altogether, these five categories encompass 556 test tasks. Detailed prompts and corresponding LLM outputs are provided in *Appendices A to* G, and an overview of the task categories is presented in Fig. 1.

3.2.1. Urban planning documentation tasks

Urban planning documentation, such as master plans, policy guidelines, and development proposals, plays a crucial role in facilitating planning decisions, coordinating stakeholders, and guiding the implementation of urban projects. During the planning process, urban planners must consider factors such as population distribution, economic layout, land use, and ecological protection, while ensuring that planning outcomes comply with regulatory policies and technical standards, coordinate with other planning types, and support broader strategic goals. This necessitates the repeated review of various related materials, official regulations, and other planning documents for the studied area, typically demanding considerable time and effort.

Our test includes three types of urban planning documentation-related tasks, each representing a key procedure involved in urban planning: 1) materials collection: given a topic or task, finding ten relevant policy documents, regulations, research papers, and other materials to assist planners in their follow-up work. 2) text information extraction: summarize the urban planning textual materials into three concise highlights. 3) urban planning documentation generation: creating coherent, relevant, and compliant content based on the given context and objectives. The generated texts should consider the conditions of the site, adhere to the formal style typical of government documents, and comply with applicable legislation. Each type contains 10 tasks, totaling 30 tasks. For materials collection tasks, the performance score is calculated as a weighted sum: it incorporates 40 % of the human-evaluated score for the collected content, 40 % of the valid materials ratio, and 20 % of the valid websites ratio. The scores of the latter two task categories are determined by human evaluators.

3.2.2. Urban planning examinations

3.2.2.1. Master of Urban Planning entrance examination. This exam serves as an admission test for undergraduate students aspiring to pursue a master's degree in urban planning. After completing three or four years of undergraduate study, students typically dedicate at least one year to preparing for this demanding and comprehensive exam. It evaluates knowledge across several key areas: 1) "The history of urban construction in China" section evaluates knowledge of the evolution of Chinese cities, from ancient slave and feudal societies to the semi-colonial, semi-feudal eras, and post-1949 urban development. 2) "The history of urban construction worldwide" section examines the progression and characteristics of urban development globally, spanning four historical periods: ancient, medieval, modern, and contemporary. 3) "The principles of urban planning" section tests foundational principles, design methods, and economic considerations in urban planning, requiring a robust grasp of both theoretical and practical knowledge. Our test of this section consists of two types of questions: multiple-choice questions (with four options) and fill-in-the-blank questions. Each type contains three categories, with 20 questions per category, totaling 120 questions.

3.2.2.2. Certified Urban-Rural Planner examination. This professional certification targets individuals with relevant academic qualifications and work experience in urban planning. The exam assesses knowledge in three primary areas: 1) principles of urban and rural planning, 2) related interdisciplinary knowledge, and 3) planning management and regulations. Compared to the master's entrance exam, this certification test places greater emphasis on understanding policies, legal frameworks, technical standards, and their practical implementation. Our test of this section consists of 60 multiple-choice questions, with 20 questions per category. Additionally, our test includes 46 open-sourced questions from Huang et al. (2024)'s C-Eval dataset. The performance score for this category is measured by accuracy, defined as the percentage of questions correctly answered by the LLMs.

3.2.3. Routine data analysis tasks

Urban planners use GIS, Python, and other tools to assess and simulate various aspects of the studied area, including its historical, current, and projected future conditions. This process requires interdisciplinary knowledge from fields such as geoscience, computer science, and environmental science. Our tasks for evaluating the data analysis assistance capability of LLMs consist of eleven task categories, totaling 271 questions (detailed in Table 1). The first ten task categories are derived directly from open-sourced datasets provided by Hochmair, Juhász, and Kemp (2024), L. Xu et al. (2024), and Hou et al. (2025). To further assess LLMs' ability to handle complex tasks, we have also included a complex GIS analysis task, which involves multiple procedures in ArcGIS.

Among these tasks, code explanation tasks are evaluated by human evaluators using accuracy, clarity, completeness, and conciseness. Accuracy refers to the factual correctness of the explanation regarding the code's functionality. Clarity refers to the ease of understanding of the explanation. Completeness refers to the extent to which the explanation covers all key aspects of the code. Conciseness refers to the efficiency of the explanation in conveying the necessary information clearly using the fewest possible words, thereby avoiding unnecessary detail. Code generation tasks and several mapping tasks are evaluated using correctness, syntax, and clarity. Correctness refers to whether the code performs its intended function as specified, without errors or unexpected results. Syntax concerns whether the code is syntactically valid, adhering to Python programming rules and free of grammatical errors or missing elements. Clarity refers to whether the code is written in a clear and understandable manner, featuring a logical structure, meaningful variable names, and appropriate comments to facilitate comprehension. Other tasks are evaluated primarily based on accuracy.

3.2.4. AI algorithm tasks

AI algorithms have already been applied across various research domains in urban studies, as outlined in the introduction. We manually compile 25 distinct AI algorithm tasks (detailed in Table 2 and *Appendix E*). The datasets used for data analysis, model training, or



Fig. 1. Five test task categories and their corresponding subtasks.

prediction—covering numeric, text, image, and spatial data—are provided in *Appendix D.*[2] LLMs are tasked with generating Python code to call the appropriate AI algorithms to complete these tasks. To evaluate the effectiveness of the Python code generated by the LLMs, we manually tested each code snippet in PyCharm. Human evaluation criteria for this category include correctness, syntax, and clarity, consistent with the evaluation standards for code generation tasks detailed in Section 3.2.3.

3.2.5. *Thesis writing task*

This section evaluates the capacity of LLMs to generate high-quality academic writing. Specifically, LLMs are asked to write an introduction of approximately 1000 words for a research paper based on the title, highlights, abstract, and research aims/questions drawn from articles in the journal *Cities*. The LLMs are also required to include relevant references in their writing. Given the increasing use of ChatGPT in academic writing, the papers selected for this test were all published before November 2022, the month ChatGPT was released. We selected five highly cited papers from diverse research domains: urban perception (Ma et al., 2021), regional economic resilience (Hu, Li, & Dong, 2022), urban agglomeration (Yaolin Liu, Zhang, Pan, Ma, & Tang, 2020), the x-minute city (Logan et al., 2022), and urban regeneration (Kim, Newman, & Jiang, 2020).

Each introduction is evaluated by human evaluators based on several criteria: whether it could be published without modifications, its readability, and the level of detail in the content. To examine the accuracy of the LLM-generated references, we manually validate the information for each reference, checking the correctness of the author names, title, publication year, journal, and other details.

4. Tests and results

4.1. *Overview*

The statistics for GPT-3.5, GPT-4o, and OpenAI o1's performance across five urban planning-related task categories are presented in Fig. 2 and Table 3. Overall, o1 generated the highest word count (80,067 words) and outperformed both GPT-4o and GPT-3.5 in all categories, achieving a superior average score of 84.08, compared to 69.30 for GPT-4o and 45.27 for GPT-3.5. These results highlight o1's exceptional performance in addressing the complex challenges of urban planning tasks.

In specific subcategories, GPT-4o demonstrated comparable performance to o1 in tasks such as code explanation, geographic feature search, and GIS concepts. However, GPT-4o outperformed o1 in material collection tasks, where it was more effective at identifying valid sources and their corresponding websites, as well as in planning management and regulations tests.

[2] https://drive.google.com/file/d/1B-lg7deUK3TxQc2NuBPNxw2fDyovOFIP/view?usp=sharing

Table 1
Overview of routine data analysis tasks.

Task	Example prompt
GIS concepts	Is the following statement true or false? The primary storage formats for spatial data include vector and raster, and these two formats can be converted between each other without loss.
Function interpretation	Provide in one sentence a common term for this equation: X = (NIR - R) / (NIR + R)
Geospatial analysis knowledge	In ArcPy, which operator would you use to compute the hyperbolic sine of a constant value? A) arcpy.sa.SinH B) arcpy.sa.Sinh C) arcpy.sa.HyperSinh D) arcpy.sa.HyperbolicSine
Dataset knowledge	What is the snippet identifier for the dataset GCOM-C/SGLI L3 Leaf Area Index (V1)? A) JAXA/GCOM-C/L3/LAND/LAI/V1 B) NASA/GCOM-C/L3/LAND/LAI/V1 C) ESA/GCOM-C/L3/LAND/LAI/V1 D) JAXA/GCOM-S/L3/LAND/LAI/V1
Toponymic identification	Papa stranded in home. Water rising above waist. HELP 812 Wood Ln, 77828. Which parts of this sentence represent location descriptions?
Geographic feature search	XX, is a city in Khatlon Region, southern Tajikistan. Located 203 km (126 mi) southeast of the capital Dushanbe on the river (a right tributary of Panj), it is one of the largest cities in the country. Return only the name of XX in the given paragraph.
Spatial reasoning	A, P, R, X, S and Z are sitting in a row. S and Z are in the centre. A and P are at the ends. R is sitting to the left of A. Show the positions of all letters from left to right.
Mapping	Create a Mapbox map with a line from Vienna to Munich.
Code explanation	Explain in one sentence the purpose of the following code: def fun(a, b): c = np.array([a[:2], b[:2]]) d = np.array([a[2], b[2]]) e = np.linalg.solve(c, d) return e
Code generation	Generate Python code that sorts the following points from East to West and prints their coordinates. Latitude,Longitude 40.7128,-74.0060 34.0522,-118.2437 51.5074,-0.1278
Complex GIS analysis task	We now have a scanned image of a section of the Mount Rainier National Park topographic map. Our task is to use this scanned image to create a 3D model for analysis and display. Task 1: Ensure that the scanned image has correct coordinate values and coordinate system. Task 2: Map glacier areas, contour lines, elevation points, and river data. Task 3: Create DEM data (20-m resolution) and produce a topographic map. Task 4: Create a 3D model of the area.

4.2. Urban planning documentation tasks

For materials collection, LLMs were tasked with identifying 100 relevant materials. As shown in Table 4, GPT-3.5's output length limitations resulted in fewer materials found, falling short of the required ten materials per task. In contrast, GPT-4o and o1 both met the requirements, with o1 providing additional resources such as reports, research papers, and guidelines beyond the initial ten. We verified the validity of materials by manually checking their titles and websites. GPT-4o achieved the highest accuracy rates for material identification (99.00 %) and website validation (90.00 %). While o1 had lower accuracy in these areas (88.80 % and 67.57 %, respectively), it offered more information, including resource overviews and task relevance assessments, generating nearly twice the word count (8233) compared to GPT-4o (4104). Key observed challenges include: 1) occasional mismatches between resource titles and website links, and 2) the inclusion of valid but less task-relevant materials.

Table 2
Overview of AI algorithm tasks.

Index	Task	Data type	Application type	Data source
1	Use K-Means to cluster Airbnb price and review score data in Singapore	Numeric	Clustering	Airbnb
2	Use DBSCAN to cluster Points of Interest (POIs) in ArcGIS Pro	Spatial	Clustering	(Yu, Yu, Wu, & Wu, 2020)
3	Use PCA to visualize the four-dimensional features of Airbnb data in two-dimensional space	Numeric	Feature extraction	Airbnb
4	Use LDA to model the topics of Airbnb comments	Text	Topic modeling	
5	Use CNN to predict house prices	Numeric	Prediction	Kaggle
6	Use LSTM to predict temperature trends	Numeric	Prediction	Kaggle
7	Use Temporal Graph Convolutional Networks to predict PM2.5 concentrations	Spatial-temporal	Prediction	Beijing Municipal Ecological and Environmental Monitoring Center
8	Use XGBoost and SHAP to identify nonlinear associations between objective attributes and subjective perceptions derived from street view images	Numeric	Association analysis	(Zhao, Lu, & Lin, 2024)
9	Use Pix2Pix GAN to generate building facades	Image	Image generation	Kaggle
10	Use a GAN to generate satellite images	Image	Image generation	Kaggle
11	Use a GAN to generate layouts of spatial elements based on given boundaries	Image	Image generation	(Yubo Liu et al., 2022)
12	Use GANmapper to create building footprints from street networks	Image	Image generation	(A. N. Wu & Biljecki, 2022)
13	Perform semantic segmentation on remote sensing images	Image	Semantic segmentation	Google Map
14	Extract building roof boundaries from remote sensing images	Image	Semantic segmentation	Google Map
15	Perform semantic segmentation of street view images based on the Cityscapes dataset	Image	Semantic segmentation	Baidu Map
16	Perform semantic segmentation of indoor images using the Segment Anything model	Image	Semantic segmentation	Google

(continued on next page)

Table 2 (*continued*)

Index	Task	Data type	Application type	Data source
17	Use YOLO to detect objects in public space images	Image	Object detection	Google
18	Estimate poses from public space images	Image	Pose estimation	Google
19	Classify street view images using the Place Pulse 2.0 dataset	Image	Image classification	(Dubey, Naik, Parikh, Raskar, & Hidalgo, 2016)
20	Conduct sentiment analysis of social media data related to museums	Text	Sentiment analysis	Dianping
21	Classify social media data from urban parks	Text	Text classification	(Zhao, Huang, Lin, & Lu, 2024)
22	Identify locations from social media data related to cultural heritage	Text	Named Entity Recognition	Dianping
23	Use BERTopic to extract topics from abstracts	Text	Topic modeling	/
24	Use BERT to summarize lengthy planning proposals	Text	Text summarization	UNESCO
25	Use GPT-4o-mini to simulate human-like agents for participatory urban planning	/	Simulation	/

https://insideairbnb.com/get-the-data
https://www.kaggle.com/code/harshwalia/house-price-advanced-regression-with-eda
https://www.kaggle.com/datasets/paulbrabban/daily-minimum-temperatures-in-melbourne
https://www.kaggle.com/code/harshwalia/house-price-advanced-regression-with-eda
https://www.kaggle.com/datasets/paulbrabban/daily-minimum-temperatures-in-melbourne

In text information extraction, GPT-3.5's performance was hindered by redundant content and complex sentence structures, often producing summaries rather than focused highlights. GPT-4o showed better organization but included excessive detail and lacked quantitative specifics. o1 achieved an optimal balance between brevity and comprehensiveness, effectively incorporating quantitative data and maintaining clear structural organization. Each highlight was distinct, contributing logically to the overall progression.

The documentation generation task is the most important in real-world applications, but even the highest-performing model, o1, only achieved a score of 74.00 (Fig. 3), indicating room for improvement. o1 provided the most thorough and detailed responses, addressing key aspects of the questions and integrating specific strategies and examples. However, three significant limitations were identified: 1) overly generic strategies that could apply to any city, 2) insufficient incorporation of city-specific data, existing policies, and regulations, reducing the specificity and persuasiveness of proposed strategies, and 3) required word count is up to 5000, while GPT-4o complied with this requirement, o1 did not. The test prompts and LLMs' responses of urban planning documentation tasks are available in *Appendix A*.

4.3. Urban planning examination tasks

The scores (Fig. 4) indicate a consistent performance improvement from GPT-3.5 to GPT-4o and OpenAI o1 across both examinations, with average scores of 39.3, 69.3, and 77.7, respectively. In the Master of Urban Planning entrance examination, o1 achieved superior scores of over 85.0 in multiple-choice questions across all three test categories, demonstrating its strong grasp of both Chinese and international urban planning knowledge. All models performed significantly better on multiple-choice questions compared to fill-in-the-blank questions, highlighting challenges in precise knowledge retrieval and contextual understanding, particularly for topics spanning ancient China, Greece, and modern urban development. For example, there are many regulation-based questions such as *'Water source protection prohibits development ( ) meters upstream and ( ) meters downstream'*. These specific figures may vary by country, and while GPT-3.5 and GPT-4o struggled to provide correct answers, o1 was able to generate accurate responses, showcasing its ability to navigate such details.

The Certified Urban-Rural Planner examination requires a higher level of expertise. The multiple-choice options in this section were often very similar, with only one or two words differing between choices, increasing the difficulty and testing the models' careful understanding of specific knowledge. As a result, o1 and GPT-4o scored 65.6 and 68.1, respectively. The test questions and LLMs' responses of urban planning examination tasks are available in *Appendix B*.

4.4. Routine data analysis tasks

As shown in Table 5 and Fig. 5, both GPT-4o and o1 showed superior performance in tasks including GIS concepts, function interpretation (such as identifying the NDVI equation), geospatial analysis knowledge (such as identifying the functions from ArcPy library), toponymic identification (extracting location data from text), geographic feature search (recognizing geographic entities and place names), and mapping (such as generating maps using Mapbox API), with average accuracies of 91.94 and 95.00, respectively. These excellent results indicate that they have a robust understanding of geospatial knowledge, with the ability to interpret and apply GIS concepts, identify and utilize geospatial functions and analytical methods, and process geographic information from various sources. This is likely due to their advanced language comprehension and data retrieval capabilities, enabling them to process and apply geospatial knowledge more effectively than GPT-3.5.

As for code explanation and generation, o1 achieved an overall score of 96.97, indicating that it can accurately interpret and write coding functions and algorithms in just a few seconds, significantly more efficient than human analysis. Specifically, code explanation is evaluated using accuracy, clarity, completeness and conciseness, and all three LLMs achieved accuracies over 93, indicating that these models can reliably understand and explain code functionality. In terms of code generation capabilities, the performance metrics reveal significant differences between models. o1 demonstrated exceptional proficiency with an average score of 96.78, notably outperforming both GPT-4o (88.00) and GPT-3.5 (84.22). Most striking was o1's near-perfect correctness score of 99.33, great syntax of 95.00, and clarity of 96.00, indicating a superior ability to use programming libraries and packages to write not just functionally but also well-structured and readable programming code for spatial data processing tasks.

In more complex analytical tasks, the performance gaps between the models became especially evident. First, spatial reasoning tasks, which required identifying spatial relationships between objects based on relative position descriptions (Hochmair et al., 2024), proved challenging for all models. Even o1 achieved only a score of 56.67, while GPT-4o (33.33) and GPT-3.5 (16.67) performed even worse. These low scores suggest that interpreting and applying spatial relationship logic remains a significant challenge for current LLMs.

Second, the complex GIS analysis task, which required LLMs to outline the step-by-step procedures in ArcMap for solving four sub-tasks in a single response, revealed noticeable performance differences. GPT-3.5's response was incomplete. While GPT-4o provided a full set of instructions, its response contained errors and lacked details needed for users to complete the tasks independently without consulting additional

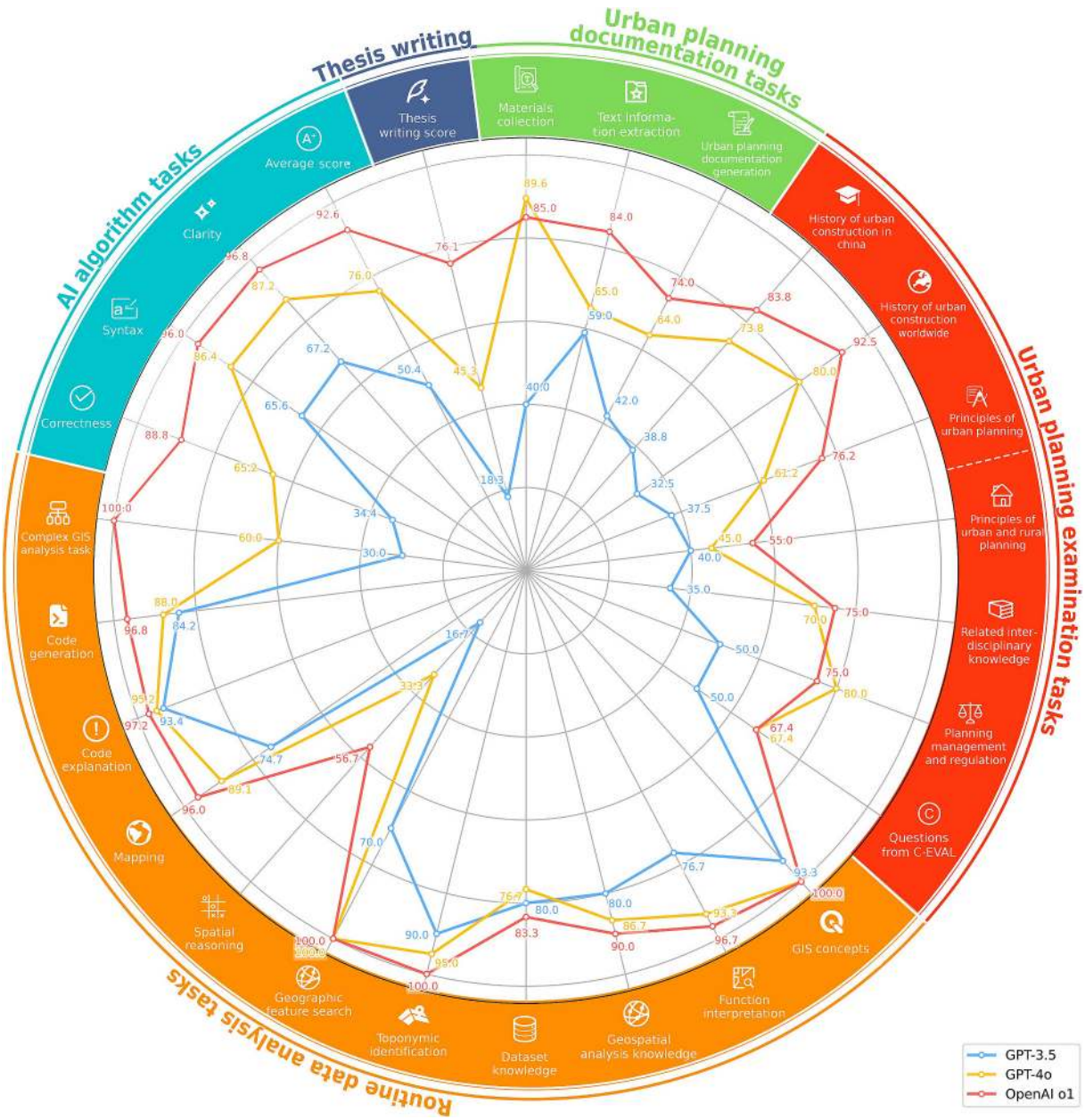


Fig. 2. LLMs' performance on urban planning-related tasks.

Table 3
Statistics of LLMs' performance on urban planning-related tasks.

	GPT-3.5	GPT-4o	OpenAI o1
Total word count	39439	51121	80067
Urban planning documentation tasks	47.00	72.87	81.01
Urban planning examination tasks	39.25	69.24	77.74
Routine data analysis tasks	71.42	83.09	92.12
AI algorithm tasks	50.40	76.00	93.40
Thesis writing tasks	18.26	45.28	76.14
Average score	45.27	69.30	84.08

resources. In contrast, o1 excelled in this task, providing a thorough and highly detailed response (2047 words), with every step meticulously outlined. For example, when providing instructions to set the data frame coordinate system, a commonly used and relatively straightforward procedure, o1's response was: *1) Right-click on **Layers** in the Table of Contents and select **Properties**. 2) Go to the **Coordinate System** Table 3) Choose **Projected Coordinate System** > **UTM** > **WGS 1984** > **Northern Hemisphere** > **WGS 1984 UTM Zone 10 N** (since Mount Rainier is between 121°W and 122°W). 4) Click **OK** to set the coordinate system.* These detailed instructions highlight o1's strength in providing comprehensive and accurate guidance in ArcGIS, making it a valuable tool for urban planners relying on GIS analysis. The test questions and LLMs' responses of routine data analysis tasks are available in *Appendix C*.

4.5. AI algorithm tasks

The complete code outputs and execution results, including both successful runs and errors, are documented in the 164-page *Appendix F*. Fig. 6 illustrates examples of successfully executed Python code outputs generated by o1. As shown in Fig. 7, o1 consistently outperformed the other models across all metrics, achieving the highest average score of 92.6, with correctness averaging 88.8, syntax 96.0, and clarity 96.8. This superior performance highlights its capacity to effectively interpret task

Table 4
Performance comparison of LLMs in urban planning documentation tasks.

Subcategory	Metric	GPT-3.5	GPT-4o	OpenAI o1
Materials collection	Generated word count	2239	4104	8233
	Number of required materials	100	100	100
	Number of found materials	78	100	125 (25 additional)
	Number of valid materials	40	99	111
	Number of provided websites	0	100	74
	Number of valid websites	0	90	50
	Score	40.00	89.60	85.03
Text information extraction	Generated word count	1330	1231	1213
	Score	59.00	65.00	84.00
Urban planning documentation generation	Required word count	Up to 5000	Up to 5000	Up to 5000
	Generated word count	2515	5020	6734
	Score	42.00	64.00	74.00
Average score		47.00	72.87	81.01

requirements, execute domain-specific operations, and provide user-friendly guidance for diverse AI algorithm applications. During our manual code evaluation, we observed several specific strengths and shortcomings of OpenAI o1, which are detailed below.

4.5.1. Key strengths of OpenAI o1

1) Both o1 and GPT-4o demonstrated competence with mature algorithms, including clustering (Task 1), dimensionality reduction (Task 3), house price prediction (Task 5), SHAP interpretation (Task 8), object detection (Task 17) and topic modeling (Task 23).

2) o1 displayed exhibited superior comprehension of diverse applications, ranging from basic unsupervised clustering to advanced machine and deep learning, effectively handling numeric, spatial, image, and text data. It effectively utilizes advanced Python libraries for targeted tasks, including Gensim (Task 4), GANs (Tasks 9, 10, and 11), pre-trained models for computer vision (Tasks 15 and 16 for semantic segmentation, Task 18 for pose estimation, Task 19 for image classification), and natural language processing (Tasks 20–24 for tasks such as sentiment analysis, text classification, named entity recognition, and text summarization). In contrast, GPT-4o, though capable, frequently generated errors or incomplete outputs, such as model usage errors (Tasks 9, 10, and 16), data format mismatches (Tasks 4, 6, 20, and 21), and poor performance on specific tasks (Tasks 11 and 15).

3) o1 excelled at executing complex, multi-step operations. In Task 25, the LLMs were tasked with instructing GPT-4o-mini to create five agents simulating residents in a renewal area. o1 successfully created the agents one by one, adhering to the prompt, while GPT-4o misinterpreted the requirements, attempting to create all five agents simultaneously with a single prompt ("You are five virtual residents..."). This led to insufficient agent discussion and substantially shorter output (1521 words) compared to o1's detailed output (3155 words).

4)Unlike GPT-3.5, which struggled with mature algorithms like GANmapper (Task 12) and Segment Anything (Task 16), o1 successfully accessed GitHub repositories to utilize documented algorithms as intended.

5)While o1 occasionally encountered minor errors, such as data format mismatches (Task 19), it quickly rectified these issues on subsequent attempts, demonstrating adaptability and reliability in complex workflows.

4.5.2. Key shortcomings of OpenAI o1

1) **Reliance on older models:** While o1 performed well with widely used models such as Pix2Pix (Task 9) and DeepLabV3 (Task 15), it struggled to leverage newer models that have replaced these older frameworks.
2) **Limited support for specialized models**: In tasks requiring less common models, such as semantic segmentation of satellite images (Task 13) or extracting building roof boundaries (Task 14), o1 defaulted to pre-trained models like ResNet or DeepLabV3+ trained on general datasets like ImageNet. These models were less effective for specialized tasks, and o1 did not make full use of relevant, fine-tuned models available on platforms like GitHub.
3) **Difficulty reproducing recent methods:** o1 faced challenges when tasked with replicating methods from recent research. In Task 2, which involved clustering POIs according to Yu et al. (2020), o1

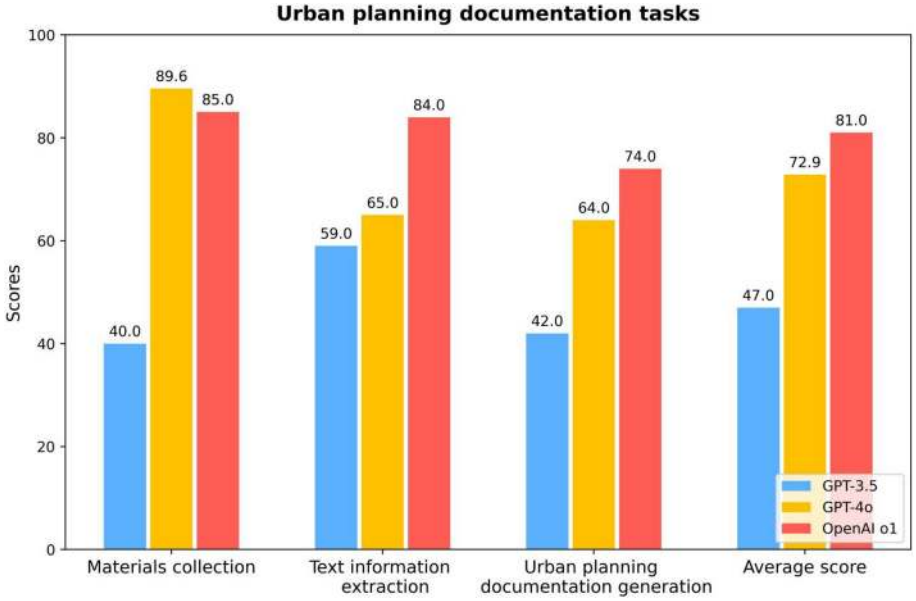


Fig. 3. LLMs' performance scores in urban planning documentation tasks.

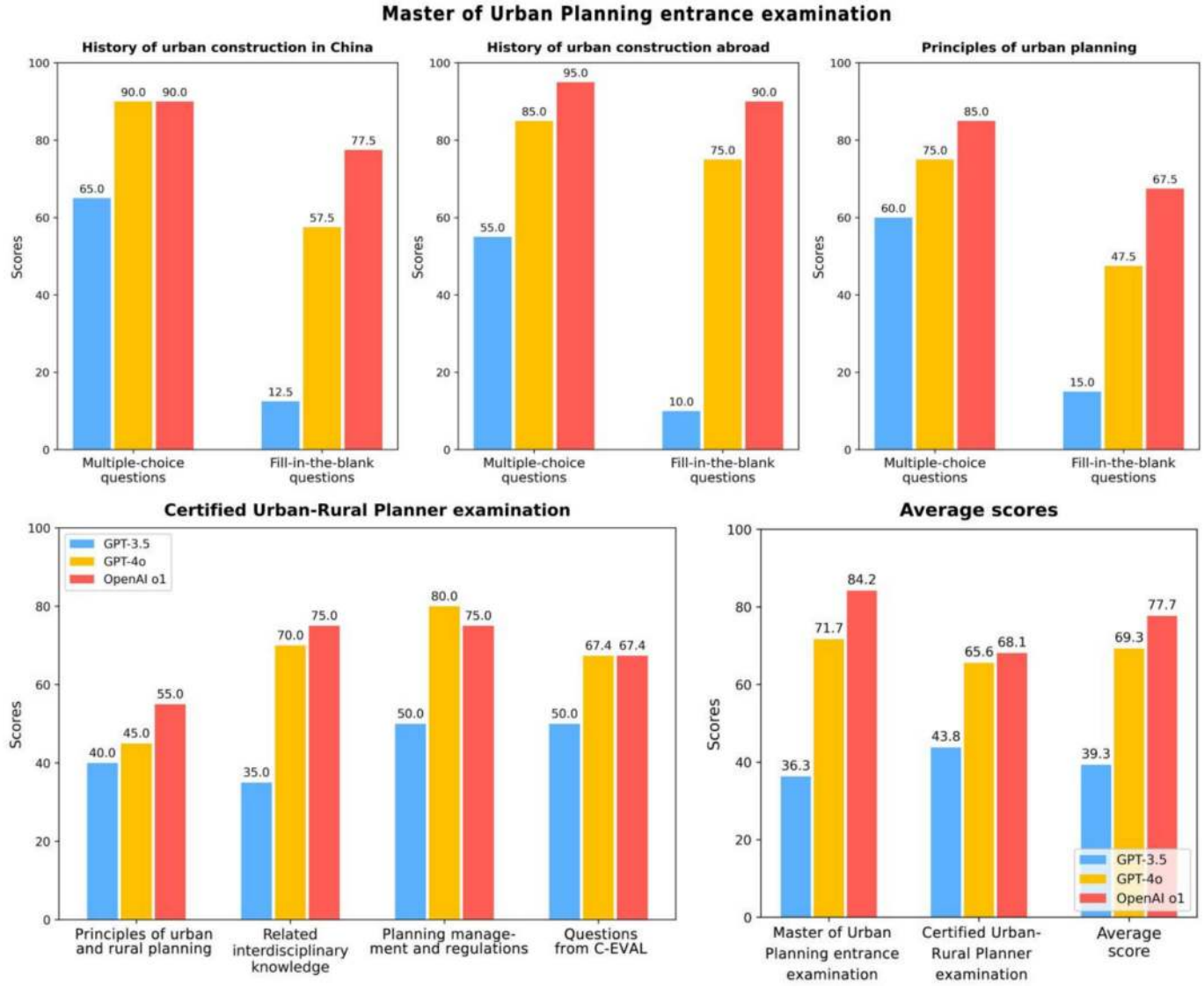


Fig. 4. LLMs' performance scores in urban planning examination tasks.

defaulted to a traditional clustering method rather than following the specified approach from the paper.

4) **Challenges with long, complex tasks:** In Task 7, which required the use of Temporal Graph Convolutional Networks to predict PM2.5 concentrations (based on Guan et al. (2024)'s study), o1 generated extensive code and guidance. However, execution errors were frequent, and fixing one issue often led to new problems in subsequent steps. This made the task difficult to complete, especially for users without domain expertise.

4.6. Thesis writing tasks

As shown in Table 6 and Fig. 8, o1 demonstrates enhanced capabilities in academic writing, consistently outperforming its counterparts across multiple dimensions. It generates introductions averaging 949.20 words, closely approximating the target length of 1000 words, while achieving moderate performance in publishable quality (63.00), readability (78.00), and adequacy (72.00). These results reflect o1's intermediate proficiency in adhering to scholarly conventions, maintaining coherent argumentation, and demonstrating better comprehension of complex research topics. It can generate overviews encompassing research contexts, study rationales, technological advancements, research gaps, and specific objectives. However, significant limitations persist in the depth and integration of literature reviews, along with inter-sentential logical coherence. Furthermore, the generated content necessitates human revision to meet strict academic standards.

Incorporating accurate and relevant references is crucial in academic writing. o1 generates the most references, totaling 111 with an average of 22.2 references per introduction, facilitating claim substantiation and literature contextualization. However, a distinct temporal disparity emerges in citation accuracy: while the model achieves 96 % accuracy for pre-2013 references, this drops significantly to 59 % for post-2013 publications. This temporal divergence reveals a critical limitation in the model's knowledge integration capabilities. While demonstrating proficiency in citing well-established sources, it shows diminished reliability in engaging with recent publications—a significant shortcoming given the pivotal role of current research in establishing the relevance and progress of a study in the introduction section. This pattern suggests potential gaps in the model's ability to accurately represent and integrate contemporary research developments, highlighting an area requiring further advancement in future iterations. The test questions and LLMs' responses of thesis writing tasks are available in *Appendix G*.

Table 5
Performance comparison of LLMs in routine data analysis tasks.

Subcategory	Metric	GPT-3.5	GPT-4o	OpenAI o1
GIS concepts	Score	93.33	100.00	100.00
Function interpretation	Score	76.67	93.33	96.67
Geospatial analysis knowledge	Score	80.00	86.67	90.00
Dataset knowledge	Score	80.00	76.67	83.33
Toponymic identification	Score	90.00	95.00	100.00
Geographic feature search	Score	70.00	100.00	100.00
Spatial reasoning	Score	16.67	33.33	56.67
Mapping	Score	74.67	89.11	96.00
Code explanation	Accuracy	95.67	97.00	97.33
	Clarity	92.33	94.33	96.67
	Completeness	91.67	91.00	96.67
	Conciseness	94.00	98.33	98.00
	Average score	93.42	95.17	97.17
Code generation	Correctness	82.00	88.33	99.33
	Syntax	88.00	95.00	95.00
	Clarity	82.67	80.67	96.00
	Average score	84.22	88.00	96.78
Complex GIS analysis task	Generated word count	217	568	2047
	Score	30.00	60.00	100.00
Average score		71.42	83.99	92.72

5. Discussion

In this section, we interpret the findings of our evaluation of OpenAI o1 in urban planning, highlighting its current capabilities alongside its persisting limitations. Drawing on these insights, we then outline prospective directions for LLM development and recommend avenues for future benchmarking research.

5.1. Significance of the study

The rapid advancement of LLMs necessitates a deep understanding of their practical applications, limitations, and potential for enhancement in specialized fields. This study makes several key contributions to the field of urban planning. Firstly, while previous review articles on AI in urban planning and urban studies (Biljecki & Ito, 2021; Chaturvedi & de Vries, 2021; Jiang et al., 2024; Koumetio Tekouabou et al., 2023) have summarized promising application directions and related research, they often stop short of evaluating how well the discussed AI algorithms actually perform in practice. Our study addresses this by providing the first comprehensive empirical evaluation of LLM performance across a wide range of urban planning tasks, moving beyond theoretical discussions to assess actual capabilities. In contrast to more extensively studied fields such as medicine (Xie et al., 2024), mathematics (de Winter et al., 2024), and education (Latif et al., 2024), urban planning lacks established benchmarks for LLM evaluation. To address this gap, we developed an open-source benchmark comprising 556 tasks across five critical categories essential to urban planning practice. This benchmark

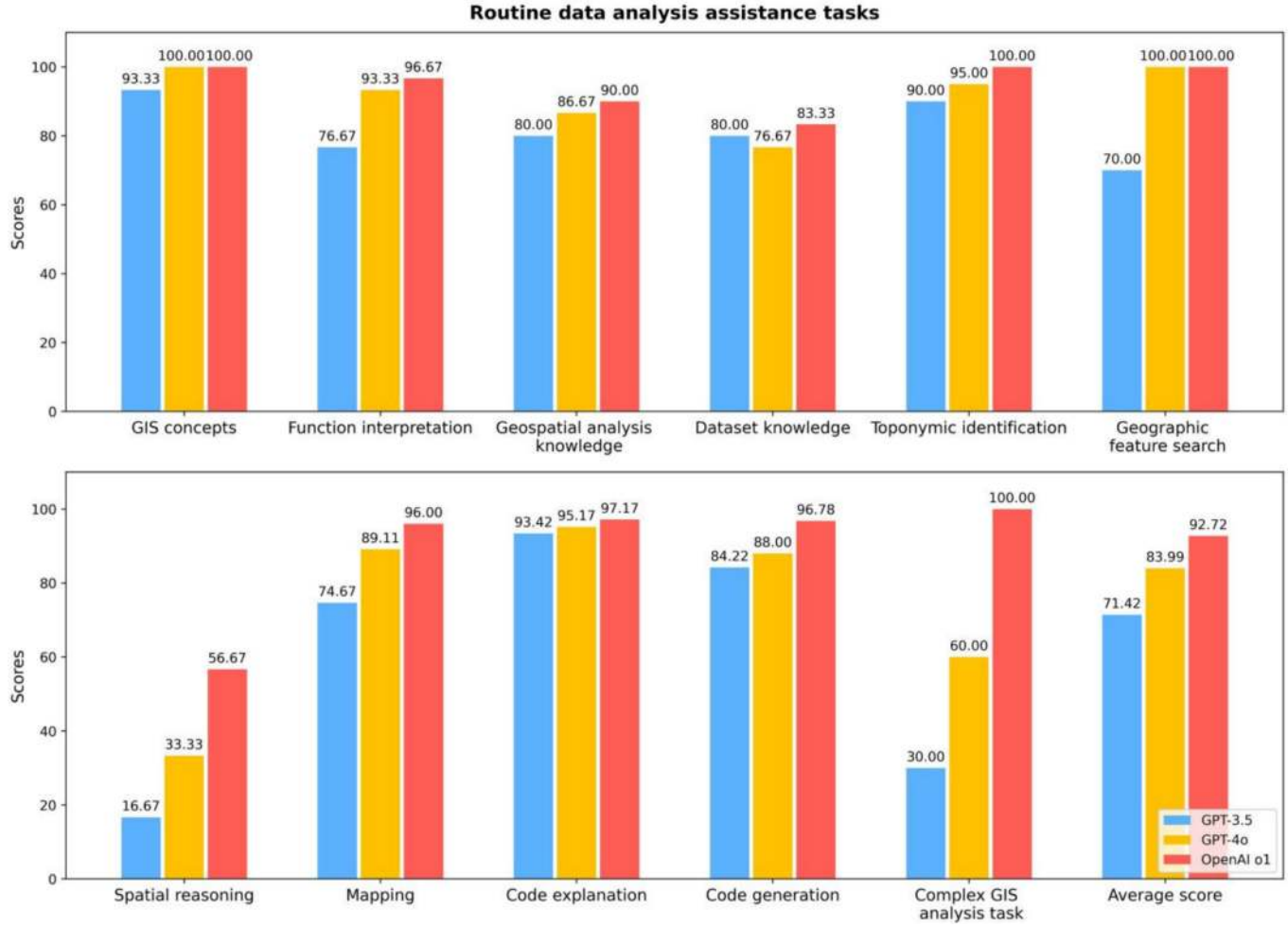


Fig. 5. LLMs' performance scores in routine data analysis tasks.

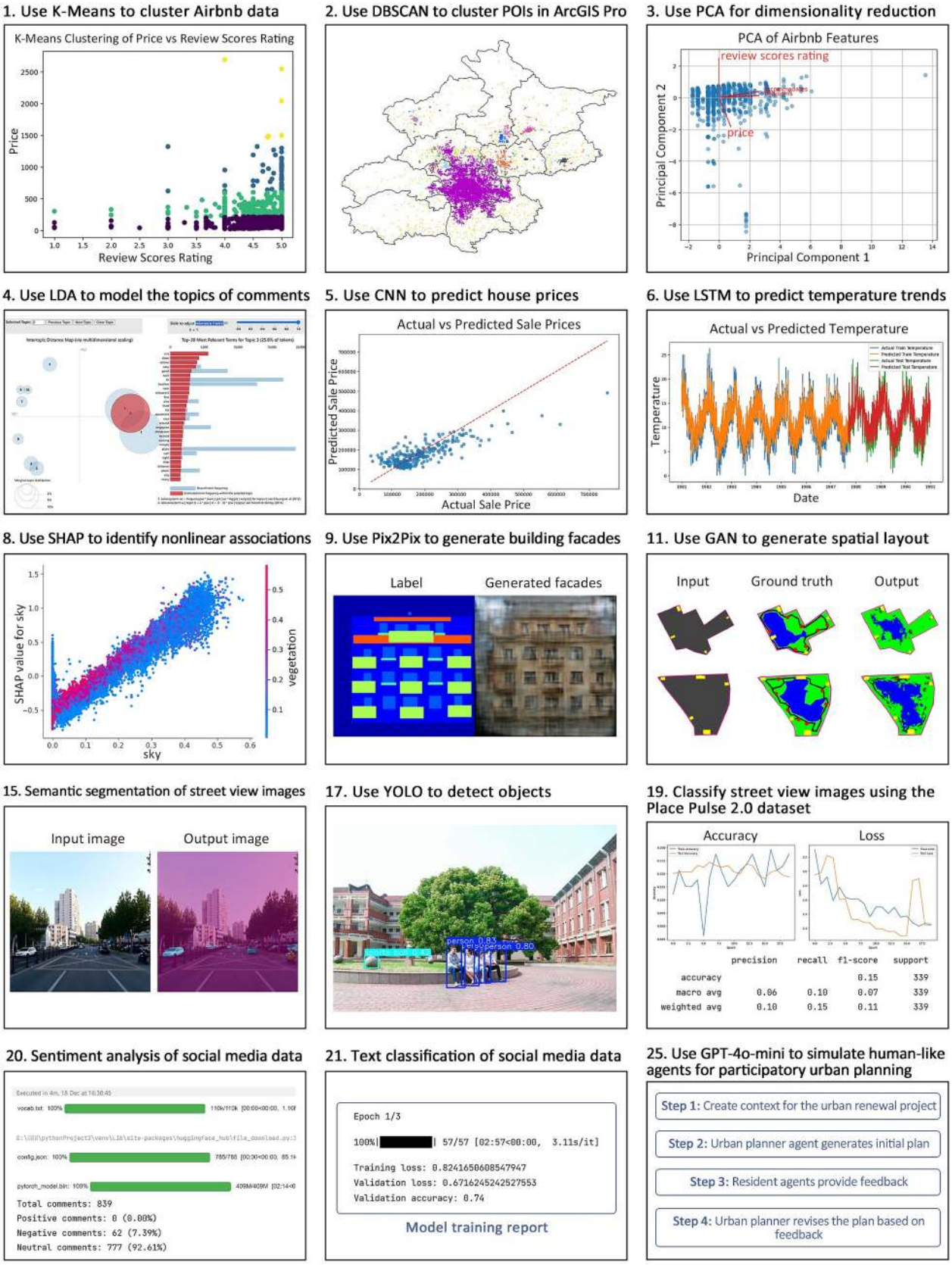


Fig. 6. Examples of successfully executed Python code outputs provided by OpenAI o1. For more details, please refer to *Appendix F*, which contains 164 pages of supplementary information.

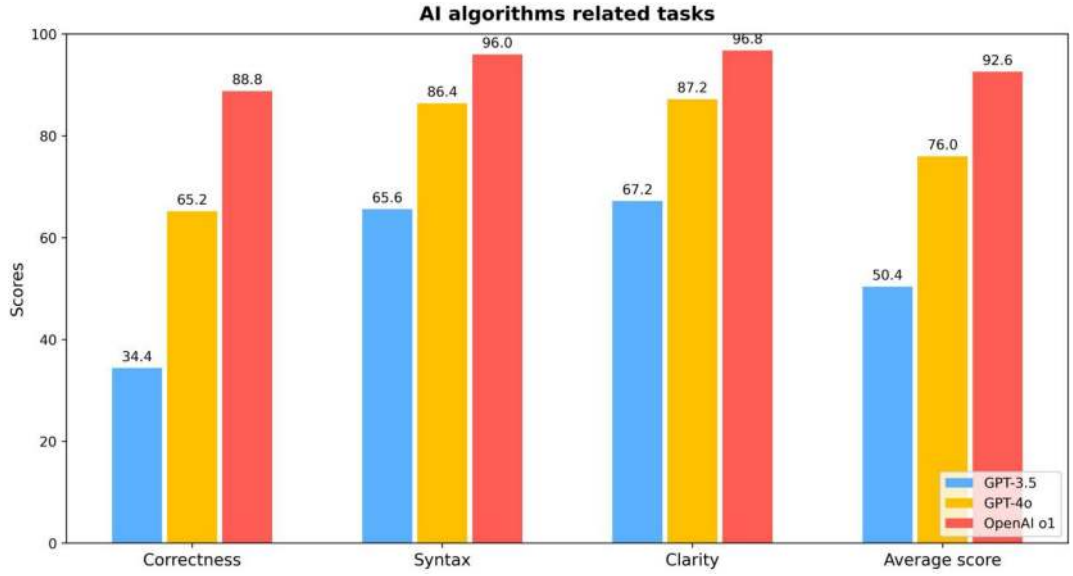


Fig. 7. LLMs' performance scores in AI algorithm tasks.

Table 6
Performance comparison of LLMs in thesis writing tasks.

Category	GPT-3.5	GPT-4o	OpenAI o1
Introduction			
Average word count	253.20	740.40	949.20
Publishable score	10.00	38.00	63.00
Readability score	38.00	58.00	78.00
Adequacy score	18.00	46.00	72.00
References			
Total number of references	0	37	111
Average references per introduction		7.40	22.20
Total accuracy	/	0.11	0.77
Accuracy of references published before 2013		0.25	0.96
Accuracy of references published after 2013		0.07	0.59
Average score	18.26	45.28	76.14

is specifically designed to assess capabilities crucial for urban planners, including foundational documentation, knowledge application, urban data analytics, and scholarly research. Unlike benchmarks created for other disciplines, ours is grounded in the unique demands of urban planning, thereby enhancing reproducibility and encouraging further research in the field.

Secondly, this study addresses the increasing integration of big data and AI algorithms in monitoring, analyzing, and predicting urban sustainable development (Son et al., 2023; F. Xu et al., 2023). While previous evaluations (Hochmair et al., 2024; Hou et al., 2025; L. Xu et al., 2024) examined various aspects of LLM performance, they overlooked AI algorithm code generation capabilities. Our research incorporates 25 typical AI algorithm tasks and provides comprehensive datasets spanning numeric, text, image, and spatial data for model training and testing.

Thirdly, the meticulous manual examination of 170,627 words of output generated by OpenAI o1, GPT-3.5, and GPT-4o ensures high reliability of the findings. This thorough evaluation, particularly critical for qualitative task results, provides professionals with detailed insights into the current capabilities and limitations of LLMs in managing domain-specific tasks. The findings significantly contribute to the emerging body of knowledge on AI integration in urban planning workflows, laying a robust foundation for future studies aimed at optimizing LLMs to better serve the specialized needs of urban planners.

5.2. Potential applications of OpenAI o1 in urban planning workflows

OpenAI o1 demonstrated a remarkable improvement over GPT-4o and GPT-3.5, achieving an average performance score of 84.08 with detailed outputs of 80,067 words, compared to 69.30 (51,121 words) and 45.27 (39,439 words) for GPT-4o and GPT-3.5, respectively. This superior performance can be attributed to o1's advanced training methodologies, including reinforcement learning for complex reasoning and pre-training on diverse, extensive datasets. o1 can produce long CoTs to refine its thinking processes, employ varied strategies, and recognize and correct mistakes before responding, thereby facilitating robust problem-solving capabilities in complex, domain-specific tasks (Jaech et al., 2024).

Our results reveal o1's particular strengths (scores over 80) in three key areas: 1) Domain knowledge mastery, including GIS concepts, function interpretation, geographic feature search, urban construction history, and geospatial analysis knowledge. 2) Basic operational competence, encompassing accurate toponymic identification, GIS task execution, mapping, materials collection and information extraction. 3) Exceptional abilities in code interpretation and generation. These competencies in natural language understanding, reasoning, and problem-solving suggest that o1 can be applied in diverse fields such as information retrieval, urban big data analytics, planning decision support (Son et al., 2023), educational assistance (Latif et al., 2024), LLM-based agent development (Zhou et al., 2024), location-based services (L. Xu et al., 2024), among others.

Given that many urban planners lack systematic education in coding, o1's advanced coding abilities offer a significant advantage. By combining exceptional code interpretation and generation with strong natural language understanding, o1 enables planners to automate complex data analysis and visualization tasks simply by describing their requirements. This capability can significantly enhance productivity and streamline urban planning workflows. Integrating such AI technologies can address complex urban big data challenges (Son et al., 2023), provide deeper insights into urban dynamics (Andrews et al., 2022), and support the evolution of urban planning practices to meet current and future urban needs (Karvonen, Cook, & Haarstad, 2020). Looking ahead, LLMs like o1 are promising to redefine workforce competencies, pushing the boundaries of human-AI collaboration in urban planning and related problem-solving domains (Latif et al., 2024). However, as an assistive tool, o1's outputs must be thoroughly verified and validated by human experts to ensure their reliability in real-world applications.

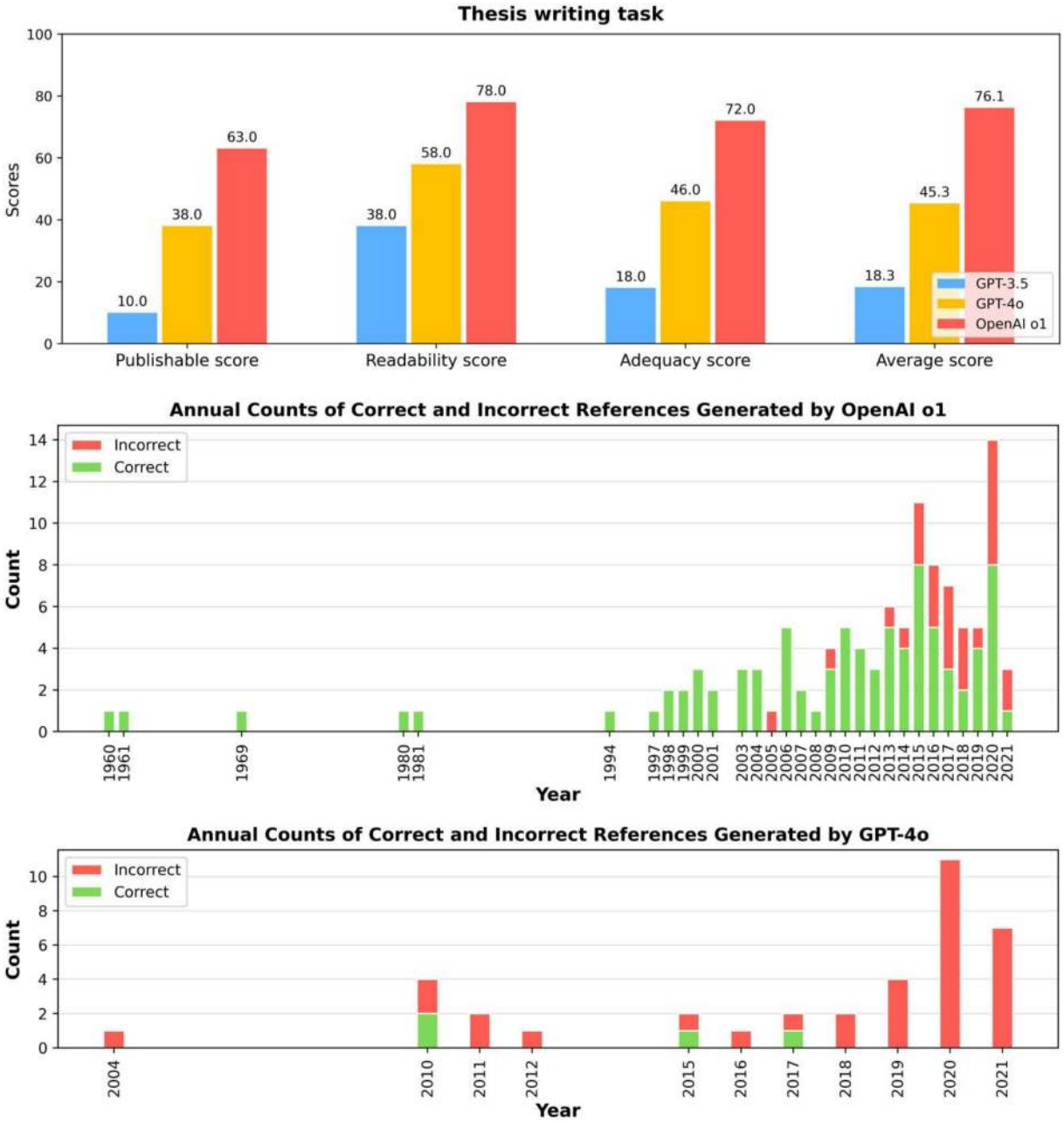


Fig. 8. LLMs' performance scores in thesis writing tasks.

5.3. Persisting shortcomings of OpenAI o1 and future improvement directions

Despite its state-of-the-art performance, o1 exhibits several shortcomings that warrant attention and future improvement.

1) **Inability in urban design:** Although o1 is a multi-modal LLM capable of integrating visual and textual inputs, it remains unable to perform urban design tasks effectively like a professional designer. This limitation in handling creative, visual, and spatial generation distinguishes urban planning from more text-centric domains and highlights a significant gap for current LLMs. Future research should prioritize enhancing multimodal integration capabilities, potentially combining with generative adversarial networks (Jiang et al., 2024; Son et al., 2023) or other effective techniques to enable more sophisticated generative urban design capabilities (Zhong et al., 2024). Another promising avenue lies in developing hybrid systems. In such systems, LLMs could act as intelligent design assistants or agents, interpreting user requirements and design constraints from natural language or sketches, and then interacting with or controlling CAD, GIS, Rhino, SketchUp, PhotoShop, and other relevant design tools to generate, modify, and evaluate urban designs. These hybrid systems combine LLMs' strengths in understanding context with the precise spatial manipulation and visualization capabilities of traditional design tools.
2) **Generation of incorrect or fabricated information** (hallucination (Liang et al., 2024; Xie et al., 2024)): As with other LLMs, a critical shortcoming observed in o1 is its susceptibility to hallucination. For instance, o1 occasionally produced invalid materials or website links in materials collection tasks, generated inaccurate references or mismatches resource details (such as titles, authors and links) and incorporated less task-relevant information in thesis writing tasks, and attempted to use non-existent code libraries in AI algorithm tasks. Beyond these instances, this inherent risk of hallucination

extends to other benchmarked urban planning tasks, threatening the reliability of planning processes and the soundness of data-driven decisions. Addressing this issue requires the implementation of robust validation mechanisms and diligent human oversight to ensure the accuracy and reliability of LLM-generated content, thereby minimizing the incidence of hallucinations.

3) **Moderate academic writing quality:** While o1 demonstrates improved academic writing capabilities, it still falls short of producing publishable, high-level content. Issues such as overly generic statements, reduced persuasiveness, redundant yet less comprehensive content, and insufficient integration of the latest research developments necessitate human revision to meet strict academic standards (Zhong et al., 2024). Incorporating specialized training modules focused on advanced academic writing and integrating up-to-date research could enhance o1's performance in this area.
4) **Incapability in high-level professional examination:** o1 underperformed in the Certified Urban-Rural Planner examination, achieving an average score of 68.1. This shortfall underscores the model's need for deeper expertise and a more refined understanding of specialized urban planning knowledge, particularly within specific regulatory frameworks. Enhancing multilingual training data quality and expanding the breadth of domain-specific knowledge in the training process could improve performance in such specialized assessments (Hou et al., 2025).
5) **Bias and regional applicability:** LLMs are trained on large datasets that often contain more information about certain regions than others. This can cause their outputs to favor planning practices and regulations common in those dominant areas, potentially making them unsuitable or lacking local relevance for urban contexts in different regions or cultures. Two key solutions involve fine-tuning the models on localized datasets and using Retrieval-Augmented Generation (RAG) to incorporate external, region-specific information.
6) **Inadequate spatial reasoning abilities:** o1 achieved a modest score of 56.67 in spatial reasoning tasks, highlighting significant challenges in complex logical reasoning and handling advanced spatial information through natural language navigation (Hochmair et al., 2024; L. Xu et al., 2024). Spatial reasoning is crucial in urban planning for arranging physical spaces effectively. Future advancements should prioritize the development of enhanced spatial reasoning capabilities, potentially through specialized training datasets (e.g., fine-tuning on extensive GIS-specific corpora rich in spatial descriptions, topological relationships, and geoprocessing instructions) and architectural innovations aimed at better integrating and reasoning over multimodal spatial inputs (e.g., maps, imagery, and vector data alongside text).
7) **Shortcomings in AI algorithm tasks:** o1 relies heavily on older, well-established models and exhibits limited support for specialized or recently developed models. Additionally, it struggles with reproducing recent research methods and managing long, complex tasks. To address these issues, future efforts should focus on integrating more comprehensive and up-to-date AI algorithms, accessing the latest models available on platforms like GitHub and Hugging Face, and implementing adaptive learning strategies that enable continuous knowledge base refinement.
8) **High operational costs**: The cost associated with using o1 is significantly higher compared to GPT-4o ($60.00 versus $10.00 per 1 M output tokens, prices as of tests conducted between October 2024 and January 2025). This cost disparity may limit widespread adoption and practical usability, especially for large-scale or resource-constrained urban planning projects.
9) **Common challenges across domains:** It is important to note that many of the identified limitations are not unique to urban planning applications. These represent common hurdles for LLMs when applied to complex, specialized, or physically-grounded domains such as robotics, engineering, medicine, and law, where precise understanding, logical deduction, and domain-specific manipulation are critical. Addressing these limitations requires ongoing foundational research in LLM architecture, training methodologies, and data curation across various specialized fields.

5.4. Limitations of the study

First, urban planning is an extensive and multifaceted field. Consequently, our study definitely did not encompass all potential LLM application areas within the discipline (Son et al., 2023). Although our experiments offer a foundational understanding of the selected LLMs' capabilities, broader insights into generalizability and robustness across more diverse urban planning scenarios would require additional research. Second, a notable limitation is that this evaluation focused exclusively on three OpenAI LLMs (GPT-3.5, GPT-4o, and OpenAI o1). Consequently, the findings may not fully generalize to the broader LLM landscape, which includes a rapidly growing number of models from other organizations, encompassing both closed-source and increasingly capable open-source alternatives. Given the rapid evolution of LLM technology, our results also may not extend to future iterations of OpenAI's models (Latif et al., 2024). Future research should therefore encompass a more diverse selection of LLMs, with a particular emphasis on leading open-source options. The open-source benchmark developed in this study offers a foundational resource for such expanded evaluations, which, especially if the benchmark is further refined, would yield a more comprehensive understanding of comparative LLM performance in urban planning and significantly advance the field.

Third, although we employed a rigorous manual examination process, certain evaluation metrics, particularly those related to qualitative assessments such as readability and adequacy in thesis writing, may be subject to reviewer subjectivity. Implementing more objective, standardized metrics or incorporating multiple reviewers could enhance the reliability of these assessments. Fourth, our study's reliance on zero-shot evaluations to establish a standardized performance baseline meant that we did not systematically investigate how different AI configurations might influence performance outcomes. Consequently, the effects of advanced prompt engineering, temperature settings, multi-turn interactions, and specialized LLM modes remain largely unexplored across the diverse urban planning tasks within our benchmark. Recognizing that optimal AI configurations vary with specific task requirements, essential future research should investigate these setting-performance dynamics. Such studies are crucial for developing invaluable practical guidelines for the effective and task-specific deployment of LLMs in urban planning.

6. Conclusions

As Popelka, Zertuche, and Beroche (2023) assert, “the real impact of AI in cities is not on the technology but on its implementation in urban planning and design.” This study reinforces this perspective, demonstrating the significant potential of advanced LLMs, particularly OpenAI o1, in enhancing urban planning processes. Our comprehensive evaluation across five key task categories, encompassing 556 tasks, reveals that o1 outperforms GPT-4o and GPT-3.5 in generating detailed documentation, excelling in examinations, assisting in data analysis, and facilitating AI algorithm implementation. These capabilities translate into improved efficiency, data-driven decision-making, and enhanced support for both academic and professional urban planning, highlighting the transformative potential of LLM integration in this field (Andrews et al., 2022).

Despite these advancements, o1 still faces challenges in addressing the complex and systemic issues cities encounter. Our study identifies several limitations, including challenges in urban design automation, susceptibility to generating inaccurate information, moderate academic writing quality, incapability to pass high-level professional examinations, inadequate spatial reasoning, limited support for specialized or

emerging AI models, and high operational costs. To fully unlock the potential of LLM in urban planning, future optimizations should prioritize enhancing multi-modal integration, implementing robust validation mechanisms to reduce errors, improving spatial reasoning, adopting adaptive learning strategies for continuous knowledge base refinement, optimizing cost-efficiency, and enabling domain-specific fine-tuning to meet the specialized needs of urban planners (Yang et al., 2024). By overcoming these challenges, LLMs can better support the evolving demands of urban planning, allowing planners to allocate more time to strategic decision-making and creative aspects. Ultimately, this could lead to the development of more sustainable, intelligent, resilient, and human-centric urban environments (Kaur et al., 2024; F. Xu et al., 2023).

From an application perspective, educators should guide students in critically evaluating and integrating AI technologies into both learning and working environments. Future research and development should prioritize the broader implementation of AI in real-world urban and infrastructure practices. It is also crucial to emphasize that the needs of urban planners should take precedence over the attributes of technology itself (Sanchez et al., 2023).

CRediT authorship contribution statement

Xukai Zhao: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **He Huang:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Tao Yang:** Supervision, Methodology, Conceptualization. **Yuxing Lu:** Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Lu Zhang:** Methodology, Conceptualization. **Ruoyu Wang:** Validation, Conceptualization. **Zhengliang Liu:** Resources, Methodology, Data curation. **Tianyang Zhong:** Resources, Methodology, Conceptualization. **Tianming Liu:** Supervision, Resources, Funding acquisition, Conceptualization.

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Declaration of competing interest

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compenvurbsys.2025.102332.

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